

Deep Learning: The Techniques and Tools You Must Know

Better late than never. If you missed the earlier discussions in this magazine related to deep learning tools and techniques, it's time to refresh your learning.

Deep learning is not a new topic as such. It's been around since 2014. However, the techniques and tools are evolving at a constant pace. A majority of them are free and open source software or FOSS. The investments are also considerable across organisations and geographies. As the practitioners say, we are in the prolonged summer phase of deep learning; so it's never too late to jump into a beautiful paradigm of DL.

AI, ML and DL

Note: DL is the new kid on the block, a subset of ML and AI.

There's always been some confusion about artificial intelligence (AI), machine learning (ML) and deep learning (DL). We often use these terms interchangeably. However, they actually are not the same. Artificial intelligence (AI) can be considered a superset of everything. AI can be viewed as automation of cognitive processes. Machine learning (ML), on the other hand, deals with the development of models out of training data sets. ML is a subset of AI. Deep learning (DL), in turn, is a subset of ML. It deals with the layering of models that have parameters, with a weight associated with each layer, and with continuously differentiable functions in a multi-dimensional space.

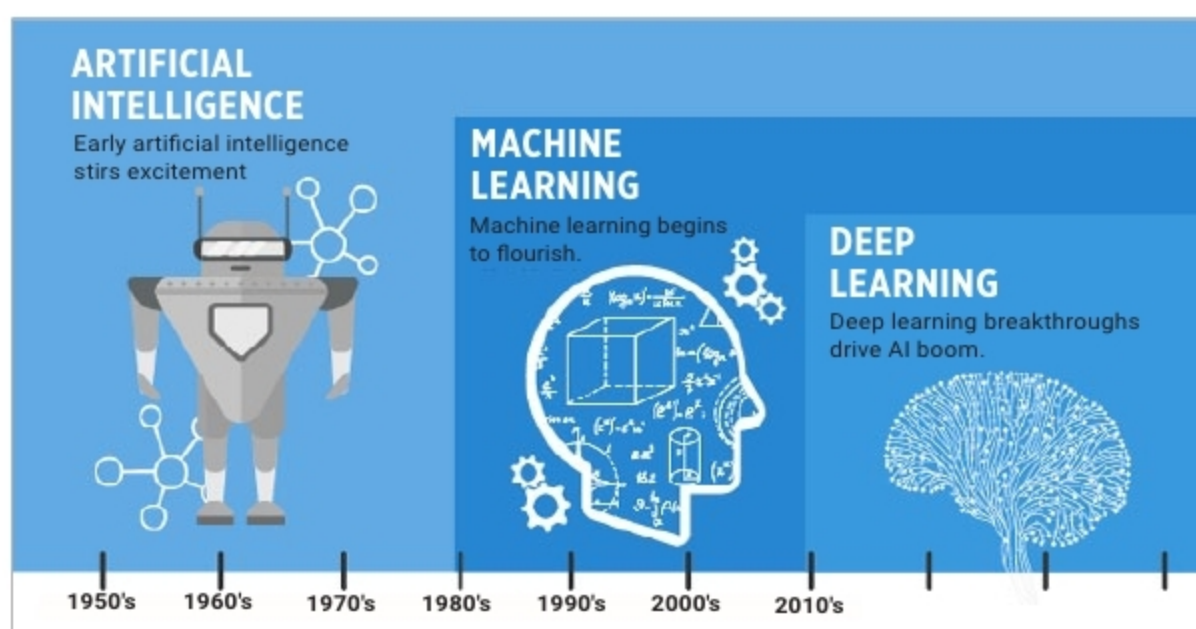


Figure 1: The relationship between AI, ML and DL (Courtesy: Nikhil Gupta, Hackernoon)

Intuition and technique

Before getting into the techniques and tools of DL, let's look at intuition first. Let us suppose that you have a crushed piece of paper in your hand that you are slowly and steadily opening out and flattening (or 'unscrambling') to its original form. DL is a similar process. When the initial data is available, it's in a scrambled and tangled form. You can hardly get anything out of it. The DL process consists of a set of deep learning layers arranged as a network of nodes, called a neural network. Each layer has a parameter that we call the weight of the layer. There can be thousands of such layers and *weights* in a typical DL neural network. The DL process is all about opening out and flattening a crushed ball of paper. By passing the 'scrambled' data (in the form of an input tensor data structure)

through the layer of the neural network, each layer is unscrambled a bit to finally flatten it out to a perfectly flat form. This final form often is in the shape of a vector, the 1D tensor.

Throughout the process, the weight of the layers is fine-tuned such that incremental unscrambling takes place. Of course, this process is iterative, and corrections over many iterations are collectively called epochs. Every time we measure the difference between what we want and what we get – the desired state minus the actual one. This gives us what is called the *loss function*. The sole purpose of the deep learning system is to minimise the loss function to attain the desired accuracy we want our DL model to achieve. This always involves a feedback loop, where the loss function is minimised in each step by calibrating the weight.

